Leiomyosarcoma (LMS) Surgical Guide

Quick Tips Care Guide

National LEIOMYOSARCOMA Foundation
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LMS Surgery: What Patients Should Know

Leiomyosarcoma (LMS) is a rare cancer that starts in smooth muscle. **Surgery is often the most important part of treatment**, and planning is highly individualized. Still, there are common principles sarcoma surgeons use when deciding what to remove and how.

The main goal of surgery is to remove the entire tumor with a rim of healthy tissue around it. Surgeons call this an **"R0 resection"**—meaning **no cancer cells are seen at the cut edge (margin)** of the removed tissue. Sometimes, removing every microscopic cell would require damaging critical nerves, blood vessels, or organs. In these situations, surgeons may leave a very small amount behind to preserve function and quality of life—often followed by **radiation or other therapy**.

Important context

- A **small positive margin** (cancer cells at the edge) **does not automatically mean worse survival**.
- Long-term outcomes are often more influenced by whether cancer **spreads to distant organs** than by tiny cells at the surgical edge.

What Surgery May Look Like

The primary goal of surgery for LMS is to remove the tumor completely while preserving as much function as possible. This often involves removing the tumor along with a margin of healthy tissue around it, known as a wide resection. This helps to ensure that all cancer cells are removed and reduces the risk of recurrence.

For LMS in the limbs or body wall:

- Surgeons aim to remove the tumor while preserving the limb and protecting nearby nerves and blood vessels when possible.
- Amputation is now rarely needed.
- If imaging suggests a very tight margin, radiation before surgery may be considered to help reduce tumor size and improve the chance of complete removal.
- If margins are positive after surgery, options may include:
  - Repeat surgery (to clear margins)
  - Radiation
  - Careful monitoring (in select cases)

For LMS in the retroperitoneum (deep abdomen):

Retroperitoneal LMS may grow large before discovery and can involve major blood vessels.

- These tumors can be challenging to remove.
- Best outcomes often occur when the tumor and any involved organs are removed together in one piece ("en bloc").
- Some tumors press on organs rather than invade them—so organ preservation may be possible in select cases.
- Avoid tumor rupture or spillage.
- Radiation before surgery usually does not reduce recurrence in this location overall, but may help in select situations when margins are expected to be close.

For LMS in the Uterus (uLMS):

Standard surgical treatment often includes:

- Hysterectomy (removal of the uterus)
- Removing the tumor in one piece
- Often removal of both ovaries, though ovarian preservation may be considered for some younger patients

Additional LMS surgical points:

- LMS rarely spreads to lymph nodes, so routine lymph node removal is usually unnecessary.
- If LMS is found unexpectedly after surgery for presumed fibroids, a second surgery may be needed to ensure complete removal and staging.
- Avoiding tumor fragmentation is critical.

Metastatic or Recurrent LMS

When LMS spreads or returns:

- If there are **only a few lung metastases**, surgery may be considered for carefully selected patients.
- **Local recurrences** may be treated with surgery when complete removal is possible.

What This Means for Patients

- **LMS surgery is specialized—experience matters.**
- Whenever possible, planning should involve a sarcoma-experienced multidisciplinary team (surgical oncology, medical oncology, radiation oncology, radiology, pathology).
- Margins matter, but so do function and quality of life.
- A small positive margin does not automatically mean a worse outcome—context matters.

Questions to Ask Your Care Team

- Is my surgery being planned or reviewed by a sarcoma specialist?
- What surgical margin is expected, and what happens if the margin is close or positive?
- Should I have radiation before or after surgery—and why?
- How will you preserve function (nerves, vessels, organ function)?
- Should I seek a second opinion at a high-volume sarcoma center?
- Can we test my tumor for genomic/molecular markers to determine whether targeted therapy is an option?

Trusted Resources

- National Leiomyosarcoma Foundation (NLMSF)
- UMass Memorial
- Clinical overview

nlmsf .org

Sarcoma & LMS Pathways

Care Guide - Quick Tips for Patients & Families

Trusted guidance for navigating diagnosis, treatment, and leiomyosarcoma

www.NLMSF.org